

1.) $\Rightarrow \vec{u}^T A \vec{v}$ je skalární s. $\Rightarrow a_{11} > 0, \det A > 0$: obměnou dostanu

$a_{11} \leq 0 \vee \det A \leq 0 \Rightarrow \vec{u}^T A \vec{v}$ není sk. součín.

1.1) $a_{11} = 0$. Bud' $A = \begin{pmatrix} 0 & b \\ c & d \end{pmatrix}$, $\vec{x} = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$, pak $\vec{x}^T A \vec{x} = (x_1, x_2) \begin{pmatrix} 0 & b \\ c & d \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} =$
 $= \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}^T \begin{pmatrix} bx_2 \\ cx_1 + dx_2 \end{pmatrix} = bx_1x_2 + cx_1x_2 + dx_2^2$. Pro $\vec{x} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \neq \vec{0}$ dostanu $\begin{pmatrix} 1 \\ 0 \end{pmatrix}^T \begin{pmatrix} 0 & b \\ c & d \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} =$
 $= b \cdot 1 \cdot 0 + c \cdot 1 \cdot 0 + d \cdot 0^2 = 0$. $\nrightarrow$ spor s 8.14 (SP)

1.2) $a_{11} < 0$. Bud' $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, $\vec{x} = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$ $\vec{x}^T A \vec{x} = (x_1, x_2) \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = x_1(ax_1 + bx_2) + x_2(cx_1 + dx_2) =$
 $ax_1^2 + (b+c)x_1x_2 + dx_2^2 \stackrel{?}{\leq} 0$. Pro $x_1 = 1, x_2 = 0$ dostanu $a \cdot 1^2 + (b+c) \cdot 0 \cdot 1 + d \cdot 0^2 = a < 0$ $\nrightarrow$
 8.14 (SP)

2.1.) $\det A = 0 \Rightarrow$ Zobrazení definované A není prosté $\Rightarrow \exists \vec{0} + \vec{v} \in \ker A \Rightarrow \vec{v}^T A \vec{v} =$
 $\vec{v}^T (A \vec{v}) = \vec{v}^T \cdot \vec{0} = 0$ $\nrightarrow$ 8.14 (SP)

2.2.) $\det A < 0 \Rightarrow \nexists B \in \mathbb{R}^{2 \times 2}$ ^{irregulární} $A \neq B^T B$ jelikož $0 < \det A \neq (\det B)^2 > 0$. Z monotónnosti $A \neq B^T B \nexists B \in \mathbb{R}^{2 \times 2}$ B reg.
 $a \geq 10.22$ (A pozitivně def. $\Rightarrow \exists B$ t.ž. $B^* B = A$) obměnou
 $\nexists B \in \mathbb{R}^2$ $B^T B \neq A \Rightarrow A$ není poz. def. $\xleftrightarrow{8.21} \vec{u}^T A \vec{v}$ není sk. součín
 $\uparrow$
 neg.

$\Leftarrow a_{11} > 0 \wedge \det A > 0$: Bud' $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$. Pak zkontrolujeme podmínky 8.14
 (Štěpán a Lukáš a čtenář)

(SSS) $\langle \vec{u}, \vec{v} \rangle = \vec{u}^T A \vec{v} = c \in \mathbb{R} = c^T = (\vec{u}^T A \vec{v})^T = \vec{v}^T A^T \vec{u} = \vec{v}^T A \vec{u} = \langle \vec{v}, \vec{u} \rangle = \langle \vec{v}, \vec{u} \rangle$

(SL1) $\langle \vec{u}, \vec{0} \rangle = \vec{u}^T A \vec{0} = \vec{u}^T (A \vec{0}) = \vec{u}^T \vec{0} = \vec{0} \cdot (\vec{u}^T A) = \vec{0} \cdot (A \vec{u}) = \vec{0} \cdot \vec{0} = 0 = \langle \vec{0}, \vec{u} \rangle$

(SL2) $\langle \vec{u}, \vec{v} + \vec{w} \rangle = \vec{u}^T A (\vec{v} + \vec{w}) = (\vec{u}^T A) (\vec{v} + \vec{w}) = (\vec{u}^T A) \vec{v} + (\vec{u}^T A) \vec{w} = \langle \vec{u}, \vec{v} \rangle + \langle \vec{u}, \vec{w} \rangle$

(SP) $\langle \vec{u}, \vec{u} \rangle = \sum \vec{u} = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = (x_1, x_2) \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = x_1[ax_1 + bx_2] + x_2[bx_1 + cx_2] = ax_1^2 + 2bx_1x_2 + cx_2^2 =$
 $a \left[\left(x_1 + \frac{b}{a}x_2 \right)^2 - \left(\frac{b}{a}x_2 \right)^2 + \frac{c}{a}x_2^2 \right] = a \left[\left(x_1 + \frac{b}{a}x_2 \right)^2 + \frac{c - \frac{b^2}{a}}{a}x_2^2 \right] = a \left[\left(x_1 + \frac{b}{a}x_2 \right)^2 + \frac{c - \frac{b^2}{a}}{a^2}x_2^2 \right] = a \left[\left(x_1 + \frac{b}{a}x_2 \right)^2 + \frac{c - \frac{b^2}{a}}{a^2}x_2^2 \right] \geq 0$
 $\uparrow$
 $a > 0$

Pokud $\vec{u} \neq \vec{0}$ tedy $x_1 \neq 0 \vee x_2 \neq 0 \Rightarrow \left(x_1 + \frac{b}{a}x_2 \right)^2 \neq 0$ nebo $x_2^2 \neq 0$. Pro $x_1 = x_2 = 0$ $\vec{0}^T A \vec{0} = 0$

$\forall$ (SSS), (SL1) a (SL2) jsme použili (4.31) a (4.29) □